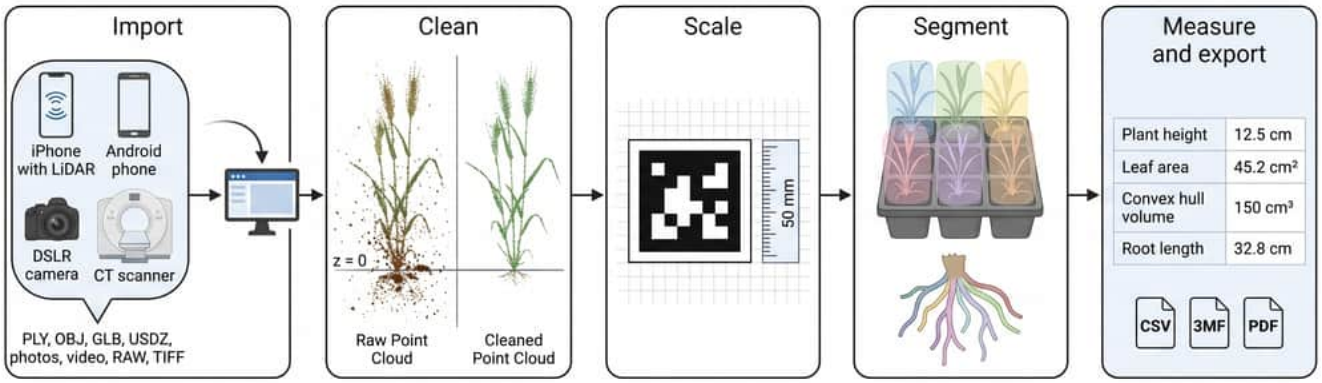
PhenoForge: from capture to traits

Which device to use, what to capture, and how the app turns it into numbers

SOP-0 · PhenoForge capture SOP · version 1.0 (October 2026)



Workflow: import → clean → scale → segment → measure and export.

Choose a capture route

Device	Best for	Metric scale	Output to PhenoForge	SOP
iPhone / iPad Pro with LiDAR	Fast single plants and trays, field work	Yes (check with markers)	PLY, OBJ, GLB, USDZ from the scanning app	SOP-1
Android phone	Low-cost capture anywhere	No: ArUco markers	App export (OBJ/GLB/PLY/splat), or video / photos reconstructed in the app	SOP-2
DSLR / mirrorless	Highest detail, repeatable lab imaging, publications	No: ArUco markers	RAW or JPEG photos reconstructed in the app	SOP-3
Any camera, excavated roots	Root crowns and washed root systems	No: ArUco or reference box	Photos → 3D roots tab	SOP-4
X-ray CT / MRI	Roots in soil, non-destructive	Yes (voxel size)	TIFF stack → Roots 3D tab	SOP-4
Motorised turntable + any camera	Repeatable lab imaging of root crowns and potted plants	No: ArUco on the platter	Session folder → Tools import (Turntable rig)	SOP-5

Illustration created in BioRender (https://app.biorender.com/illustrations/a8401c1320ac7f644c979d35). Screenshots: PhenoForge (MIT licence).

Rules that apply to every device

Do

- Put the ArUco card flat beside every plant, in view from every side.
- Use diffuse, even light; avoid wind (plants must not move during capture).
- Use a plain, matte, non-green background.
- Record plant ID, genotype and treatment in the library (or a QR label in view).
- Capture a colour chart photo at the start of each session for comparable greenness.

Avoid

- Direct sun and hard shadows, glossy pots and reflective surfaces.
- Fast movement, digital zoom, HDR or beauty filters on phones.
- Changing focus or zoom between shots (photogrammetry needs fixed optics).
- Trusting one number without a check: validate against manual measurements (Validation tab).

Print the ArUco sheet from **Tools > Printable ArUco marker sheet** (DICT_4X4_50, 50 mm). Print at 100 %, measure one marker with a ruler, and write the measured size on the card. Glue it flat on rigid card; a curled marker reads short.

Analyse in PhenoForge

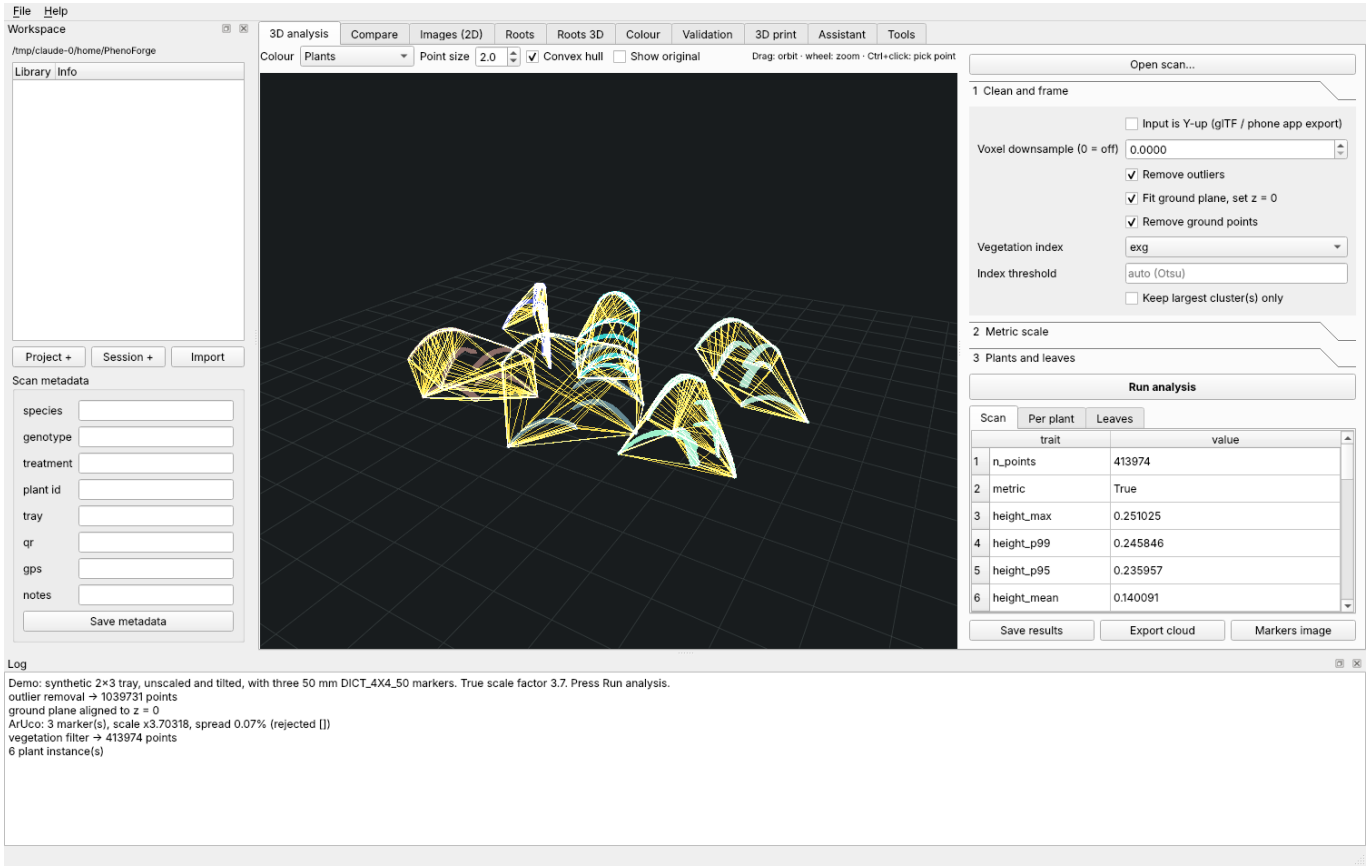
1 **Open the scan**
3D analysis tab > **Open scan…**, or add it to the library (left panel: Project + / Session + / Import) so metadata and results are stored with it.

2 **Clean and frame**
Keep *Remove outliers*, *Fit ground plane* and *Remove ground points* ticked. Tick *Input is Y-up* for GLB/glTF/USDZ exports from phone apps if the plant lies on its side.

3 **Set the scale**
Section 2: choose **ArUco markers** and type the measured marker side. For LiDAR scans choose *Already metric* but still check: **Markers image** shows each marker's size after scaling.

4 **Segment**
Section 3: *single* for one plant, *watershed* for a tray of touching plants (enter the number of plants), or *grid* with rows, columns and pot pitch for a bench of equally spaced pots.

5 **Run and export**
Press **Run analysis**. Results appear in the Scan / Per plant / Leaves tables; **Save results** writes JSON and CSV. Colour the view by Height, Greenness or Plants to check the segmentation.



3D analysis tab after running the demo tray: six plants separated, convex hulls shown, ArUco scale within 0.1 %.